BME 6145 Midterm Prediction Examination

Applied Artificial Intelligence for Biomedical and Healthcare Applications

I. Multiple Choice Questions

1. Which description best distinguishes machine learning from an expert system?
A. It learns patterns from data
B. It always uses rules by hand
C. It never needs training
D. It cannot make predictions
Correct answer: A.
Explanation: Machine learning estimates patterns from examples rather than relying only on hand-coded rules.

2. After training, a model applied to an unseen patient is performing:
A. labeling
B. inference
C. tokenization
D. normalization
Correct answer: B.
Explanation: Inference is the use of learned parameters on new inputs.

3. Which trend is associated with modern AI scaling?
A. Less data and less compute
B. Rules replacing data
C. More data, parameters, and compute
D. Removing validation
Correct answer: C.
Explanation: Modern systems commonly scale training data, model parameters, and computational resources.

4. Predicting blood glucose from BMI is an example of:
A. classification
B. clustering
C. generation
D. regression
Correct answer: D.
Explanation: Blood glucose is a continuous target, so the task is regression.

5. Predicting Type 1 versus Type 2 diabetes is:
A. classification
B. regression
C. dimensionality reduction
D. reconstruction
Correct answer: A.
Explanation: The target consists of discrete categories.

6. Unsupervised learning uses:
A. only test labels
B. unlabeled inputs
C. hand-written rules
D. a fixed answer key
Correct answer: B.
Explanation: Unsupervised methods discover structure without target labels.

7. In a Jupyter notebook, execution order matters because:
A. cells cannot contain code
B. Markdown changes labels
C. cells share kernel state
D. imports are forbidden
Correct answer: C.
Explanation: Variables and imports persist in the active kernel and depend on prior execution.

8. The main purpose of a Conda environment is to:
A. replace all notebooks
B. remove dependencies
C. label training data
D. isolate packages and interpreters
Correct answer: D.
Explanation: Environments reduce conflicts by isolating project-specific Python versions and packages.

9. For linear regression, the model is commonly written as:
A. $\hat{y} = wx + b$
B. $\hat{y} = \sigma(w^T x)$
C. $\hat{y} = \max(0, x)$
D. $\hat{y} = x^2 + b^2$
Correct answer: A.
Explanation: A linear model uses a weighted input plus an intercept.

10. The standard regression loss emphasized in the course is:
A. accuracy
B. MSE
C. AUC
D. cross-entropy only
Correct answer: B.
Explanation: Mean squared error measures squared prediction errors for continuous outputs.

11. In $E^{(i)} = \hat{y}^{(i)} - y^{(i)}$, $E^{(i)}$ is the:
A. learning rate
B. bias parameter
C. prediction error
D. class label
Correct answer: C.
Explanation: It is the difference between prediction and observed target for sample i.

12. Gradient descent updates parameters in the direction:
A. of increasing cost
B. of larger errors
C. of random noise
D. opposite the gradient
Correct answer: D.
Explanation: The negative gradient points toward decreasing cost locally.

13. If the learning rate is too large, training may:
A. diverge or oscillate
B. stop needing data
C. guarantee a minimum
D. remove the bias
Correct answer: A.
Explanation: Large steps can overshoot the minimum.

14. For a suitable learning rate in basic linear regression, the cost generally:
A. increases forever
B. decreases toward a minimum
C. becomes a label
D. ignores parameters
Correct answer: B.
Explanation: Gradient descent follows downhill directions on the convex cost surface.

15. Logistic regression is mainly used for:
A. image compression
B. clustering only
C. binary classification
D. DNA tokenization
Correct answer: C.
Explanation: It maps a linear score to a probability for binary classification.

16. The sigmoid function maps a real-valued score to:
A. $(-\infty, \infty)$ only
B. integers only
C. image pixels
D. $[0, 1]$
Correct answer: D.
Explanation: $\sigma(z) = 1/(1 + e^{-z})$ produces values between zero and one.

17. Binary cross-entropy strongly penalizes:
A. confident wrong predictions
B. correct uncertain predictions only
C. large images
D. short sequences

Correct answer: A.
Explanation: Log loss becomes large when a model assigns high probability to the wrong class.

18. A decision boundary from logistic regression is linear in:
A. the training epochs
B. the input features
C. the labels only
D. the validation set
Correct answer: B.
Explanation: The thresholded linear score defines a hyperplane in feature space.

19. Standardization is useful before neural-network training because it:
A. creates labels
B. guarantees no overfitting
C. aligns feature scales
D. replaces testing
Correct answer: C.
Explanation: Similar scales can improve optimization and prevent one feature from dominating numerically.

20. A biological neuron is abstracted mathematically as:
A. clustering plus PCA
B. token plus position
C. loss plus dataset
D. weighted sum plus activation
Correct answer: D.
Explanation: The artificial neuron computes $z = w^T x + b$ and then applies an activation.

21. A perceptron can represent:
A. linearly separable classes
B. every nonlinear relation
C. only regression
D. only unlabeled data
Correct answer: A.
Explanation: A single perceptron has one linear decision boundary.

22. Which logic gate motivates hidden layers?
A. AND
B. XOR
C. OR
D. NOT only
Correct answer: B.
Explanation: XOR is not linearly separable by one straight boundary.

23. The main purpose of a hidden layer is to:
A. remove all labels
B. prevent forward propagation
C. learn nonlinear feature transformations
D. sort patients
Correct answer: C.
Explanation: Composed nonlinear transformations permit nonlinear decision boundaries.

24. Forward propagation computes:
A. gradients only
B. test labels
C. PCA directions
D. predictions from inputs
Correct answer: D.
Explanation: Inputs pass through layers to produce an output prediction.

25. Backpropagation uses the chain rule to compute:
A. parameter gradients
B. patient diagnoses
C. token IDs
D. PCA scores only
Correct answer: A.
Explanation: It propagates loss derivatives backward through the network.

26. The usual hidden-layer activation in the genomics notebook is:
A. sigmoid only

B. ReLU
C. softmax only
D. identity only
Correct answer: B.
Explanation: ReLU helps optimization and reduces saturation compared with sigmoid in hidden layers.

27. A major issue with saturated sigmoid units is:
A. too many labels
B. no output range
C. vanishing gradients
D. loss of all inputs
Correct answer: C.
Explanation: Sigmoid derivatives become very small at extreme inputs.

28. The output activation for binary classification in the notebook is:
A. ReLU
B. PCA
C. convolution
D. sigmoid
Correct answer: D.
Explanation: A sigmoid output represents a binary-class probability.

29. The loss paired with the binary sigmoid output is:
A. BinaryCrossentropy
B. MSE only
C. PCA loss
D. categorical distance
Correct answer: A.
Explanation: Binary cross-entropy is appropriate for binary probabilistic predictions.

30. In a supervised dataset, the response variable is:
A. the optimizer
B. the target label
C. the batch size
D. the kernel width
Correct answer: B.
Explanation: The response is the outcome the model learns to predict.

31. One-hot encoding DNA bases represents each base as:
A. a continuous diagnosis
B. a loss value
C. a four-dimensional indicator vector
D. a PCA axis
Correct answer: C.
Explanation: Each of A, C, G, and T receives one indicator dimension.

32. A DNA binding-site classifier in the notebook predicts whether a sequence:
A. contains an image
B. has high glucose
C. is a PCA component
D. binds a transcription factor
Correct answer: D.
Explanation: The simulated assay labels sequences as binding or non-binding.

33. A validation set is mainly used to:
A. monitor generalization during training
B. update every weight directly
C. replace training data
D. encode nucleotides
Correct answer: A.
Explanation: Validation performance helps detect overfitting and select training choices.

34. If training accuracy rises while validation accuracy falls, this suggests:
A. under-sampling only
B. overfitting
C. PCA compression
D. perfect generalization
Correct answer: B.
Explanation: The model is fitting training examples without improving unseen-data performance.

35. Batch size is the number of samples used to:
A. define the test label
B. create a token vocabulary
C. estimate one gradient update
D. calculate PCA axes
Correct answer: C.
Explanation: A batch supplies examples for one optimizer update.

36. A CNN kernel primarily detects:
A. only global averages
B. class labels directly
C. environment conflicts
D. local patterns
Correct answer: D.
Explanation: Convolution scans local patches for learned motifs.

37. CNNs reduce parameters through:
A. local connections and weight sharing
B. larger dense layers
C. removing activations
D. using no training
Correct answer: A.
Explanation: The same kernel is reused across local regions instead of using a separate weight everywhere.

38. A convolution output at one position is commonly a:
A. global label count
B. local dot product
C. PCA eigenvalue only
D. token ID
Correct answer: B.
Explanation: The kernel weights are multiplied with a local input patch and summed.

39. In microscopy, early CNN layers may detect:
A. clinical outcomes directly
B. whole patients
C. edges and lines
D. validation splits
Correct answer: C.
Explanation: Deeper layers combine simple local features into shapes and morphology.

40. The receptive field is the input region that can:
A. define the optimizer
B. store labels
C. replace a kernel
D. influence one unit
Correct answer: D.
Explanation: It describes which input positions are accessible to a feature unit.

41. Compared with CNNs, self-attention can connect distant tokens in:
A. one direct hop
B. no hops
C. only one local pixel
D. a separate dataset
Correct answer: A.
Explanation: Each token can attend directly to every other token.

42. Tokenization converts a sequence into:
A. gradients
B. discrete tokens
C. PCA components
D. loss curves
Correct answer: B.
Explanation: Words, subwords, nucleotides, or k-mers can serve as tokens.

43. A DNA language model may tokenize a sequence using:
A. only diagnoses
B. only images
C. bases or k-mers
D. loss values

Correct answer: C.
Explanation: Both single nucleotides and fixed-length substrings are possible.

44. Learned token embeddings are:
A. binary labels only
B. gradients only
C. test predictions only
D. dense vectors
Correct answer: D.
Explanation: An embedding layer maps token IDs to trainable vector representations.

45. Positional encoding is needed because attention alone:
A. does not preserve order
B. cannot use vectors
C. always labels data
D. removes sequence length
Correct answer: A.
Explanation: Position information is added to make otherwise order-insensitive attention sequence-aware.

46. In self-attention, Q and K primarily determine:
A. training labels
B. relevance weights
C. convolution width
D. PCA variance
Correct answer: B.
Explanation: Query-key similarity produces attention scores.

47. In self-attention, V supplies:
A. the learning rate
B. the target label
C. information being mixed
D. the test split
Correct answer: C.
Explanation: Values are combined using the attention weights.

48. The scaled dot-product attention formula contains:
A. $MSE(w, b)$ only
B. $wx + b$ only
C. $\text{ReLU}(x)$ only
D. $\text{softmax}(QK^T/\sqrt{d_k})V$
Correct answer: D.
Explanation: This is the course formula for self-attention.

49. Causal masking in autoregressive generation prevents a token from seeing:
A. future tokens
B. its own embedding
C. past tokens
D. its position
Correct answer: A.
Explanation: Next-token prediction must use only the prefix.

50. Autoregressive generation repeatedly:
A. removes all tokens
B. predicts and appends the next token
C. calculates PCA first
D. changes labels
Correct answer: B.
Explanation: The predicted token is appended and becomes part of the next prefix.

51. PCA finds directions of:
A. minimum sample count
B. maximum label error
C. maximum projected variance
D. random token order
Correct answer: C.
Explanation: Principal components summarize directions with large variance.

52. PCA is primarily a:
A. binary classifier

B. language tokenizer
C. gradient optimizer
D. linear dimensionality-reduction method
Correct answer: D.
Explanation: PCA uses linear combinations of input features.

53. The first principal component explains:
A. the greatest variance
B. no variance
C. all labels
D. only noise
Correct answer: A.
Explanation: PC1 is chosen to maximize projected variance.

54. A PCA score is:
A. a class label
B. a sample coordinate in PC space
C. a kernel weight
D. a batch index
Correct answer: B.
Explanation: Projection onto PCs gives low-dimensional sample coordinates.

55. An autoencoder consists of:
A. tokenizer, label, test
B. kernel, bias, label only
C. encoder, bottleneck, decoder
D. PCA, AUC, batch
Correct answer: C.
Explanation: The encoder compresses and the decoder reconstructs the input.

56. The latent space of an autoencoder is:
A. a raw label table
B. a validation metric
C. a kernel image
D. a compressed representation
Correct answer: D.
Explanation: The bottleneck stores lower-dimensional learned features.

57. A standard autoencoder is trained to:
A. reconstruct its input
B. predict only future tokens
C. maximize class count
D. remove all features
Correct answer: A.
Explanation: The input serves as the reconstruction target without requiring labels.

58. A limitation of standard PCA is that it may miss:
A. all linear variation
B. nonlinear curved structure
C. sample coordinates
D. correlated features
Correct answer: B.
Explanation: PCA represents linear directions and may poorly capture branching nonlinear trajectories.

59. The notebook metric AUC is useful for evaluating:
A. Python syntax
B. token length only
C. ranking of binary predictions
D. PCA reconstruction only
Correct answer: C.
Explanation: AUC summarizes discrimination across classification thresholds.

60. Which split should remain untouched for final performance estimation?
A. training set
B. batch set
C. token set
D. test set
Correct answer: D.
Explanation: The test set estimates generalization after model choices are finalized.

II. True/False Questions

61. Machine learning always requires hand-written rules.
Correct answer: False.
Explanation: It can learn patterns from data.

62. Inference uses a trained model on new inputs.
Correct answer: True.
Explanation: Inference follows training.

63. Regression predicts a categorical label by definition.
Correct answer: False.
Explanation: Regression predicts a continuous output.

64. Unsupervised learning requires correct target labels.
Correct answer: False.
Explanation: It discovers structure without target labels.

65. A Jupyter notebook can contain code and Markdown cells.
Correct answer: True.
Explanation: Both are standard cell types.

66. MSE is commonly used for continuous regression targets.
Correct answer: True.
Explanation: It averages squared prediction errors.

67. Gradient descent moves in the direction of increasing cost.
Correct answer: False.
Explanation: It moves opposite the gradient.

68. A very large learning rate can cause oscillation.
Correct answer: True.
Explanation: Updates may overshoot the minimum.

69. The sigmoid output is always below zero.
Correct answer: False.
Explanation: It lies between zero and one.

70. Cross-entropy is suitable for binary probabilistic classification.
Correct answer: True.
Explanation: It penalizes incorrect probabilities.

71. A single perceptron can solve XOR with one linear boundary.
Correct answer: False.
Explanation: XOR is not linearly separable.

72. Hidden layers can create nonlinear decision boundaries.
Correct answer: True.
Explanation: Layered nonlinear transformations enable this.

73. Backpropagation computes gradients using the chain rule.
Correct answer: True.
Explanation: It propagates derivatives through layers.

74. ReLU is defined as $\max(0, z)$.
Correct answer: True.
Explanation: Negative inputs map to zero.

75. Standardization guarantees that a model cannot overfit.
Correct answer: False.
Explanation: It helps optimization but does not guarantee generalization.

76. One-hot encoding assigns one active base dimension per DNA nucleotide.
Correct answer: True.
Explanation: A, C, G, and T use four indicator dimensions.

77. The validation set should directly update model weights in ordinary supervised training.
Correct answer: False.
Explanation: It monitors choices rather than providing training updates.

78. A growing training score with falling validation score is consistent with overfitting.
Correct answer: True.
Explanation: The model is losing unseen-data performance.

79. Batch size is the number of classes.
Correct answer: False.
Explanation: It is the number of samples per update.

80. CNN kernels use the same learned weights at different positions.
Correct answer: True.
Explanation: This is weight sharing.

81. CNNs connect every input pixel to every hidden unit by definition.
Correct answer: False.
Explanation: Convolution uses sparse local connections.

82. A receptive field describes accessible input positions.
Correct answer: True.
Explanation: It is the input region influencing a unit.

83. Self-attention can directly connect distant sequence positions.
Correct answer: True.
Explanation: Every token can compare with every other token.

84. Positional encoding is unnecessary for ordered sequences.
Correct answer: False.
Explanation: Attention needs position information to represent order.

85. Query-key similarity helps determine attention weights.
Correct answer: True.
Explanation: QK^T produces relevance scores.

86. PCA requires class labels to find principal components.
Correct answer: False.
Explanation: PCA is unsupervised.

87. PC1 captures the greatest projected variance.
Correct answer: True.
Explanation: That is its defining optimization criterion.

88. PCA can perfectly represent every nonlinear biological trajectory with one component.
Correct answer: False.
Explanation: PCA is linear and may miss curved structure.

89. An autoencoder uses its input as a reconstruction target.
Correct answer: True.
Explanation: This permits label-free representation learning.

90. A positive result in melanoma recurrence prevention proves prevention in healthy people.
Correct answer: False.
Explanation: That clinical population and endpoint are different.

III. Short-Answer Questions

91. Distinguish AI, machine learning, neural networks, deep learning, and generative AI.
Reference answer key: AI is the broad goal; ML learns from data; neural networks are trainable layered functions; deep learning uses many layers; generative AI produces new samples.

92. Explain supervised learning and give one biomedical regression and one classification example.
Reference answer key: Supervised learning uses inputs with target labels. Predicting glucose is regression; predicting diabetes type is classification.

93. State the gradient-descent workflow for linear regression.
Reference answer key: Initialize w, b; compute predictions and cost; calculate gradients; update w, b opposite the gradient; repeat until stopping.

94. Why can a perceptron solve AND but not XOR?
Reference answer key: AND is linearly separable by one hyperplane. XOR has alternating classes, so no single straight boundary separates them; hidden layers are needed.

95. Describe forward propagation and backpropagation in an MLP.
Reference answer key: Forward propagation computes predictions layer by layer. Backpropagation propagates the loss gradient backward using the chain rule, then parameters are updated.

96. Explain why CNNs are parameter-efficient for images or genomic motifs.
Reference answer key: CNNs use sparse local connections and shared kernels. They reuse the same detector across positions and learn hierarchical local features.

97. Compare CNNs and self-attention for long-range sequence relationships.
Reference answer key: CNN receptive fields grow through stacked local layers. Self-attention connects distant tokens directly in one layer, although it can be computationally expensive.

98. Describe tokenization, embedding, and positional encoding in a Transformer.
Reference answer key: Tokenization creates discrete units; embeddings map IDs to learned dense vectors; positional encoding adds order information before attention.

99. Explain PCA and one limitation in biomedical data analysis.
Reference answer key: PCA finds linear directions of maximum variance and projects samples into fewer coordinates. It may miss nonlinear curved or branching structure.

100. Describe the personalized mRNA cancer-vaccine workflow and the role of AI.
Reference answer key: Tumor sequencing identifies neoantigens; AI ranks likely immunogenic targets; selected sequences are delivered as mRNA in lipid nanoparticles to stimulate T cells.

IV. Long/Essay Questions

101. Explain why AlexNet was a turning point for modern deep learning and connect its success to CNN principles.
Reference answer key: Discuss ImageNet-scale labeled data, GPU training, and CNN architecture. Explain hierarchical features, local connections, weight sharing, and fewer parameters than dense networks. Note that deeper CNNs still face long-range-dependency limitations.

102. Design a reliable workflow for training a binary DNA-binding classifier using the provided notebook concepts.
Reference answer key: Encode bases with one-hot encoding; split into training, validation, and test sets; train a network with ReLU hidden units and sigmoid output; use BinaryCrossentropy and an optimizer such as Adam; monitor accuracy/AUC and validation behavior; evaluate once on untouched test data and avoid leakage.

103. Compare logistic regression, a single perceptron, and an MLP for biomedical classification.
Reference answer key: All compute weighted sums and activations at a basic level. Logistic regression uses a sigmoid and linear boundary; a perceptron is a single error-trained neuron with linear separation; an MLP adds hidden nonlinear transformations and can represent XOR-like boundaries but may overfit.

104. Explain how a Transformer represents and generates a biological sequence.
Reference answer key: Tokenize bases or k-mers, map IDs to embeddings, add positional information, and form Q, K, V. Self-attention mixes values using query-key scores; causal masking enables next-token prediction, and generation appends predictions iteratively.

105. A gene-expression study has thousands of features and limited samples. Compare PCA with an autoencoder and recommend an analysis strategy.
Reference answer key: PCA gives an unsupervised linear maximum-variance projection and interpretable variance summaries. An autoencoder learns encoder–latent–decoder nonlinear representations by reconstruction. Standardize appropriately, inspect explained variance/reconstruction and visualizations, validate downstream use, and avoid treating latent structure as causal evidence.